= Natural Language Professing =

Pro. Dr. Nurul Huda

Date: 11 July 2026 (1st class)

CT : 4th class [open book] ⟹ 25 mark [20 marks = CT, 5 marks = Assignment]

Mid Exam: No Exam ⟹ 1/0 Assignment (Group 5 members) ⟹ 30 marks

↓

to be submitted before final Exam

Report + Assignment PPT

↓

individual

Attendance ⇒ 5 marks

final Exam ⇒ 40 marks (As per schedule of UIU)

100 marks

NLP :

Human Language ⟺ Computer/Device

↑

computer Science

Artifical Intelligence

computer Linguistics

NLP TOOLS :

30 marks

1. Speech Recognition (STT) : for hearing impaired peop
   ↳ Speech to text – Dragon, Siri, Amazon Alexa, Cortor
2. Speech Synthesis (TTS) : for Blind peoples
   ↳ Text to Speech – Ivona, Natural reader, ispeech, voice Reader 15.

3. Machine Translation : for foreigner - google translator, Yandex Translate, systarn, Bing translator
4. Question Answering System (QAS) — google, chatgpt, START, BERT

Assignment : NLP tools (1)

5 Sentence/voice প্রত্যেকের ৩টি free Softwar নিতে হবে।

- 3টি সহজ
- 2টি কঠিন/লম্বা Sentence

Noice add করতে হবে // machine sound
Surroundings Sound
vehicle movement sound

Languag: Bangla
Hindi
Arabic
English
} any three(3) Language

কি কি কারণে voice Recognition fail হতে পারে তার একটি ব্যাখ্যা

Noice
Gender
Speaking rate
dialogue differences

NLP TOOLS (2)

প্রত্যেক text to speech (type করে window তে input করা)

## NLP TOOLS (3)

Idioms ~~বা~~ Phrase ব্যবহার করতে হবে। like – cats and dogs

Block, Bank, Bed of Roses

নিখুত দিতে হবে (text)

Note: NLM MODEL

## NLP tools (4)

Confusing type Question and lets find out the accuracy of answer.

- ☐ Speepch Recognition কি?
- ☐ কিভাবে কাজ করে? Importance/demerits
- ☐ Problem গুলো কি?
- ☐ Example (5 sentence) গুলি উল্লেখকরা
- ☐ Conclusion

## Introduction

1. STT
2. TTS
3. MT
4. QAS

## BASIC

Alphabet = $\sum$

Bangla Alphabet = { অ, .... ঔ, ক, ..... ড়, ঢ়, ৎ, ং- }

English Alphabet {A, ..... Z, a, ... z}

Binary Alphabet {0, 1}

## String

Bangla Strings over Bangla Alphabet

Strings = { ঐ, ও, বই, বউ, বক, .... কলম, কলম, ফুল }

English Strings over English Alphabet

Strings: { I, a, .... an, In, to, ..... ant, Pen, .... }

## Length of String

|কলম| = 3

|বই| = 2

|Pen| = 3

|ফুল| = 3

|অক্ষম| = 3 (ব্যঞ্জনবর্ণ)

## Concatenation of Strings

x = চাঁদ

y = মুখ

xy = চাঁদমুখ

yx = মুখচাঁদ

## Power of Alphabet

$\sum^1 = \{A, \ldots Z, a, \ldots z\}$

$\sum^0 = \{\epsilon\}$

→ Empty String $|\epsilon| = 0$

$\Sigma^1 = \{I, a, \ldots\}$

$\Sigma^2 = \{am, an, at \ldots\}$

$\Sigma^3 = \{ant, pen \ldots\}$

$\vdots$

$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots\ldots$

$= \{\epsilon\} \cup \{I, a, \} \cup \{am, an, at \ldots\}$

$\cup \{ant, pen, \ldots\} \ldots\ldots$

$= \{\epsilon, I, a, am, at, an \ldots ant, pen \ldots\}$

$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots = \{I, a\} \cup \{am, an, at \ldots\}$

$\cup \{ant, pen \ldots\}$

$= \{I, a, am, an, at \ldots ant, pen \ldots\}$

$\therefore \Sigma^+ = \{I, a, am, an, at \ldots ant, pen \ldots\}$

$= \{\epsilon, I, a, am, an, at \ldots ant, pen \ldots\} - \{\epsilon\}$

$= \Sigma^* - \{\epsilon\}$ $\quad \boxed{\Sigma^+ = \Sigma^* - \{\epsilon\}}$

# DFA (Deterministic Finite Automate)

Draw a DFA that supports all integer numbers over alphabet $\{+, -, 0, \ldots, 9\}$

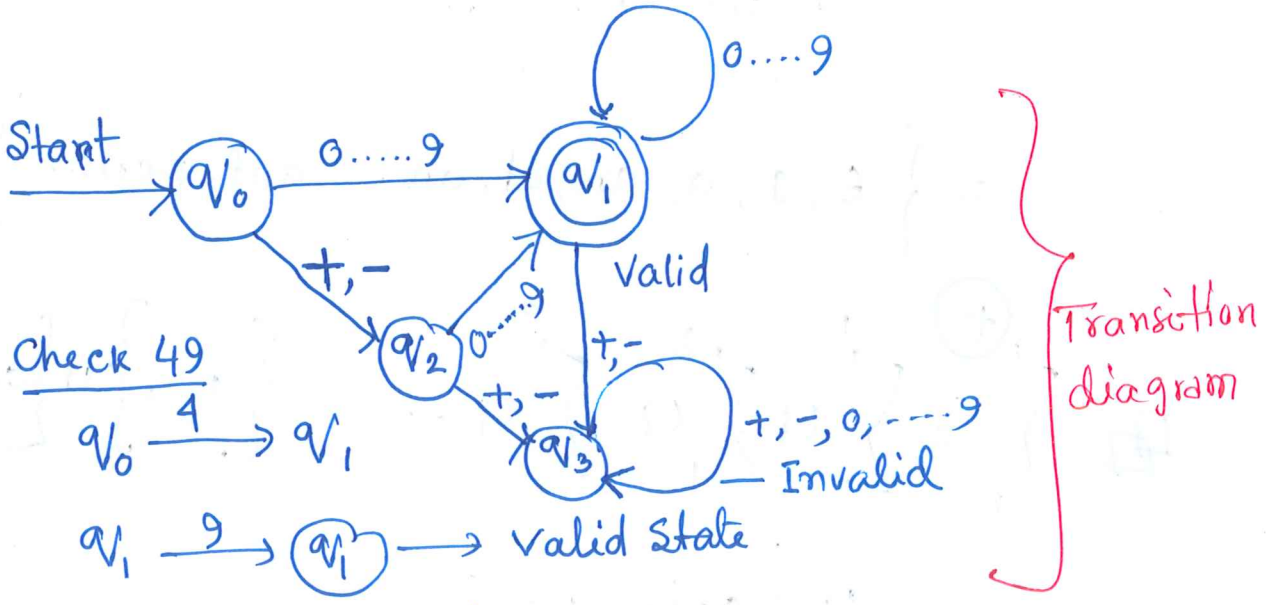
Example,

| Valid: | | | |
|---|---|---|---|
| | 2 | +2 | -2 |
| | 9 | +9 | -9 |
| | 23 | +23 | -23 |
| | 49 | +49 | -49 |

Invalid:

23-9

+-239




Check 49

$q_0 \xrightarrow{4} q_1$

$q_1 \xrightarrow{9} (q_1) \rightarrow$ Valid State

Given 49 valid String

Check +93

$q_0 \xrightarrow{+} q_2$

$q_2 \xrightarrow{9} q_1$

$q_1 \xrightarrow{3} (q_1) \rightarrow$ Valid State

Check +-93

$q_0 \xrightarrow{+} q_2$

$q_2 \xrightarrow{-} q_3$

$q_3 \xrightarrow{9} q_3$

$q_3 \xrightarrow{3} (q_3) \rightarrow$ Invalid State

Check -93+

$q_0 \xrightarrow{-} q_2$

$q_2 \xrightarrow{9} q_1$

$q_1 \xrightarrow{3} q_1$

$q_1 \xrightarrow{+} (q_3) \rightarrow$ Invalid State

Sample Question: MSCSE ID check করার DFA কিভাবে করবো।

[Date 18/7/2026 2nd class]

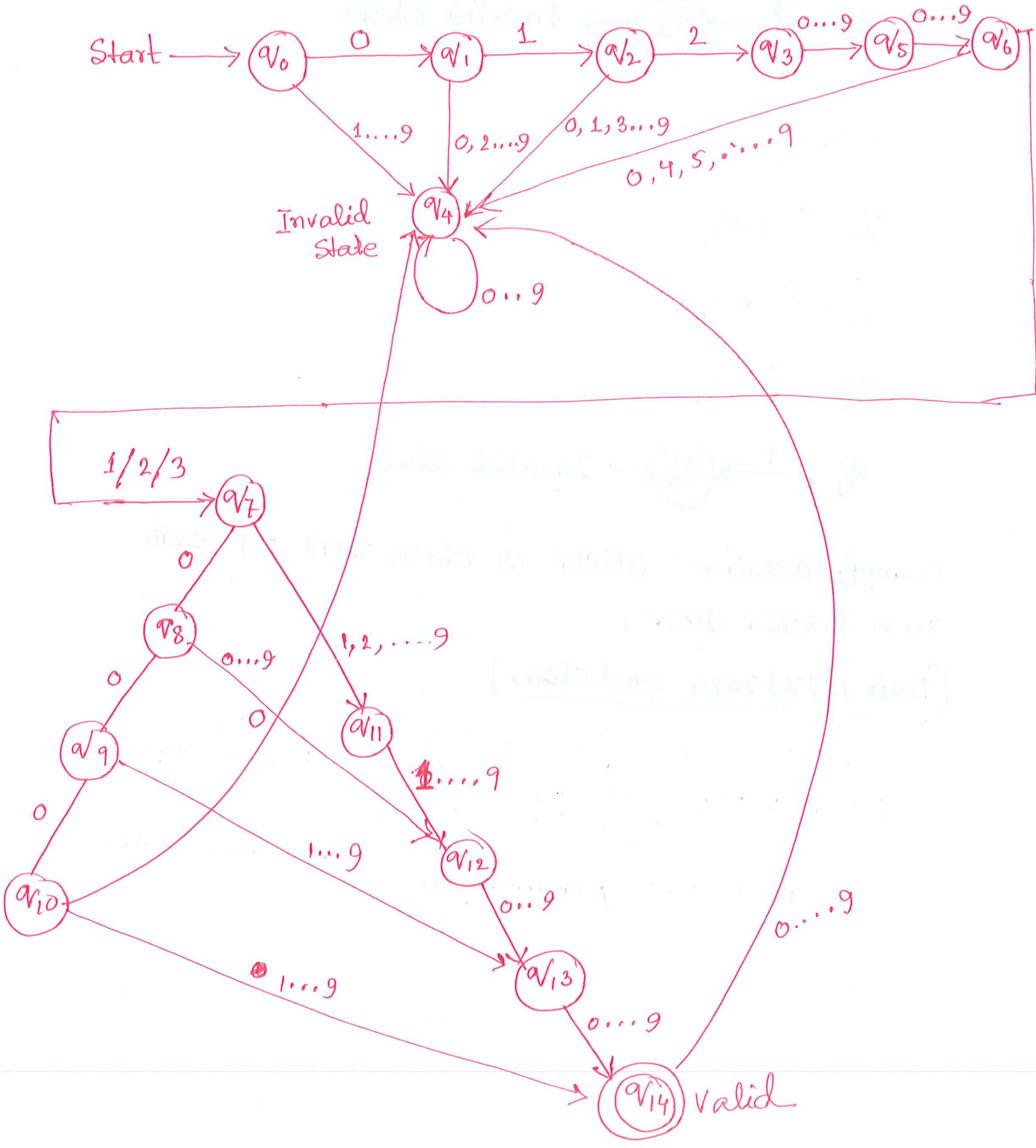
Construct a DFA from MSCSE ID of UIU
Trimester spring-1, summer-2, Fall-3

For Example 012 23 2 0023

012 → MSC code
23 → Admission year
2 → Trimester
0023 → Roll (must not be zero)

# DFA

- Starting State ($q_0$)
- Final State / Accepting State
- Transition



0122610002 check

$q_0 \xrightarrow{0} q_1$

$q_1 \xrightarrow{1} q_2$

$q_2 \xrightarrow{2} q_3$

$q_3 \xrightarrow{2} q_5$

$q_5 \xrightarrow{6} q_6$

$q_6 \xrightarrow{1} q_7$

$q_7 \xrightarrow{0} q_8$

$q_8 \xrightarrow{0} q_9$

$q_9 \xrightarrow{0} q_{10}$

$q_{10} \xrightarrow{2} \boxed{q_{14}}$ ⇒ valid state

0122610002 0

$q_0 \xrightarrow{0} q_1$

"

"

"

"

"

"

"

"

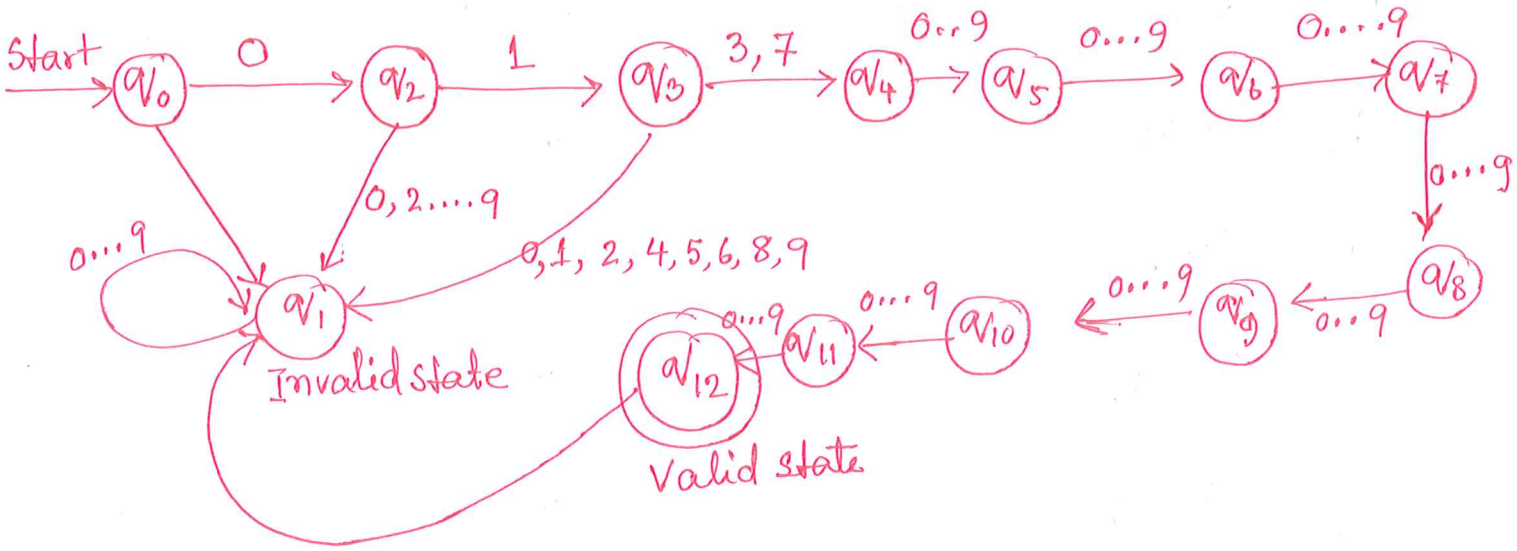
$q_{10} \xrightarrow{2} q_{14}$

$q_{14} \xrightarrow{0} \boxed{q_4}$ Invalid state

Grameen phone no:

017.....

013.....




DRAW A DFA for decimal numbers

1

23

239

+1

-23

-239

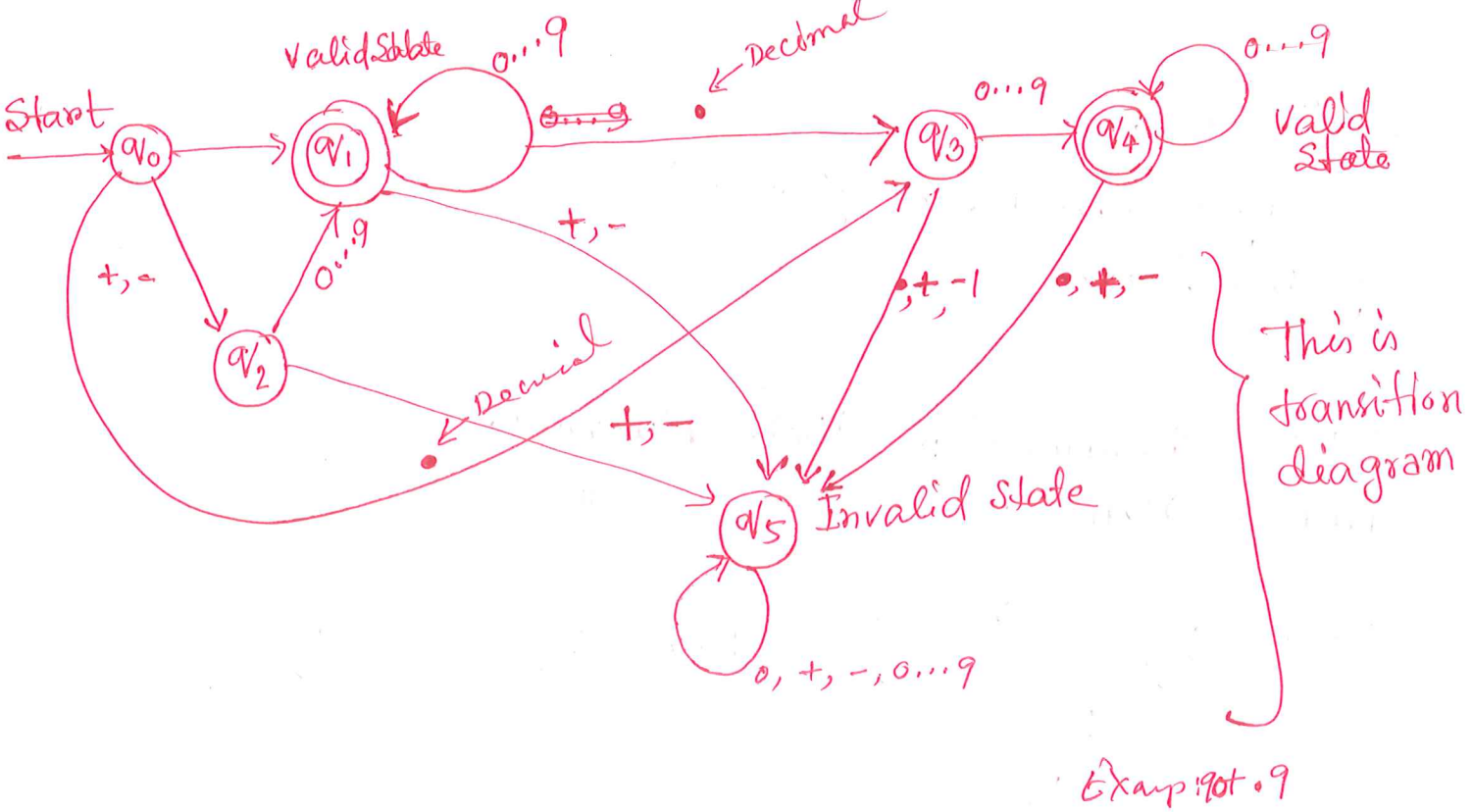
23.9

-23.4

+23.0

-23.091

.56




Example of .9

** Valid এর ক্ষেত্রে যখন Double লাইন Circle এবং Invalid এর ক্ষেত্রে কিন্তু Single Line Circle হবে।

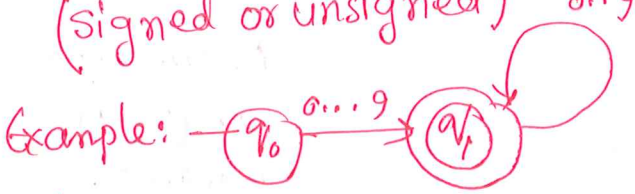
*** সবগুলোর Alphabet = { 0...9, +, -, . }

## TRANSITION TABLE

| | 0....9 | +, - | . |
|---|---|---|---|
| $q_0$ | $q_1$ | $q_2$ | $q_3$ |
| $q_1$ | $q_1$ | $q_5$ | $q_3$ |
| $q_2$ | $q_1$ | $q_5$ | $q_5$ |
| $q_3$ | $q_4$ | $q_5$ | $q_5$ |
| $q_4$ | $q_4$ | $q_5$ | $q_5$ |
| $q_5$ | $q_5$ | $q_5$ | $q_5$ |

** এগুলোর diagram যেমন হবে এবং Question হবে which language this DFA support?

Ans: This DFA supports all decimal numbers (signed or unsigned)

Example: 


↓

all positive Integers

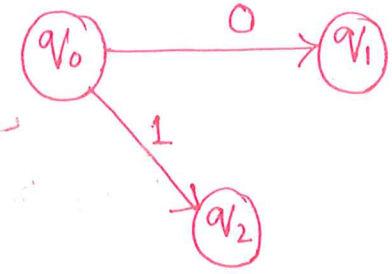
## WHY DFA?

- Information retrival from text
- Web Search

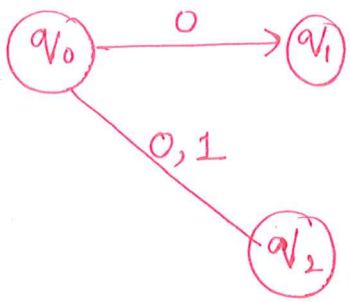
DFA — Deterministic finite Automate

NFA — non " " "

DFA




NFA




### Exercise:

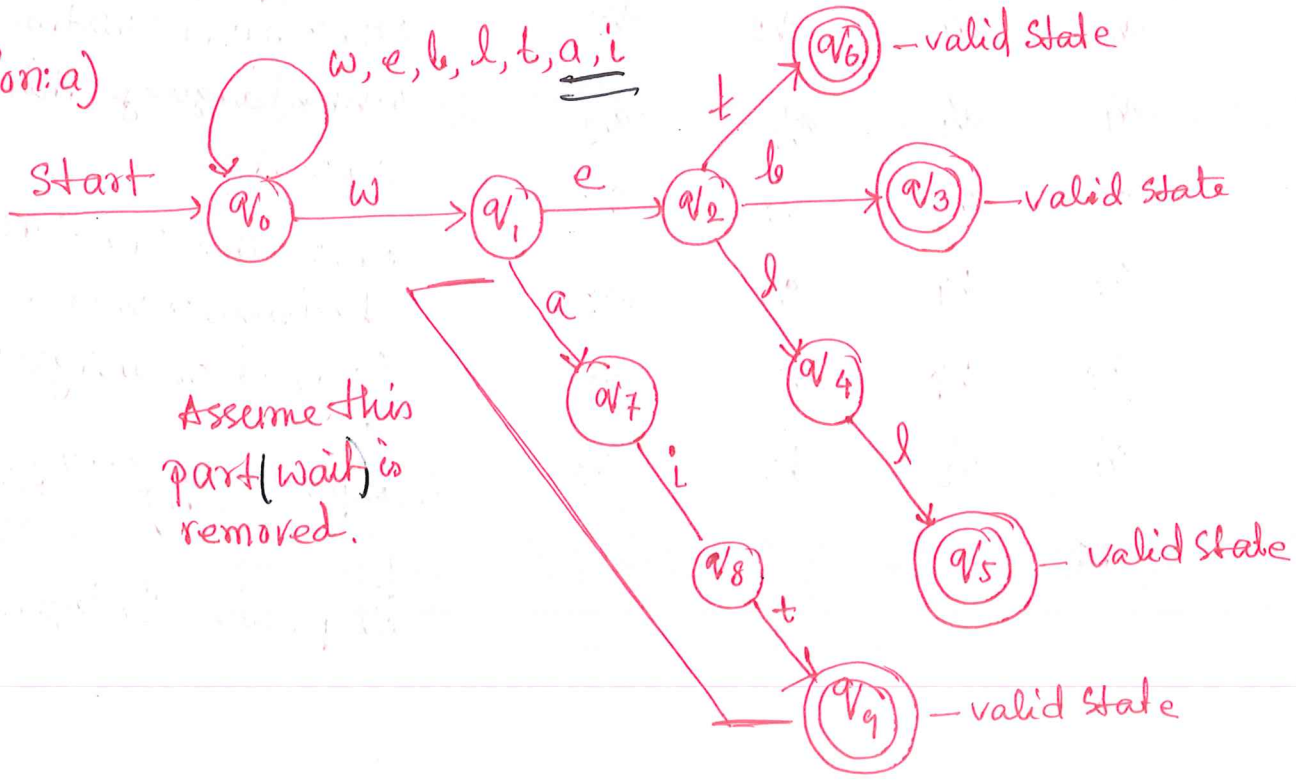
a) Draw an NFA to support the following English words in web search over Alphabet { w, e, b, l, t, a, i }

Assume it is removed.

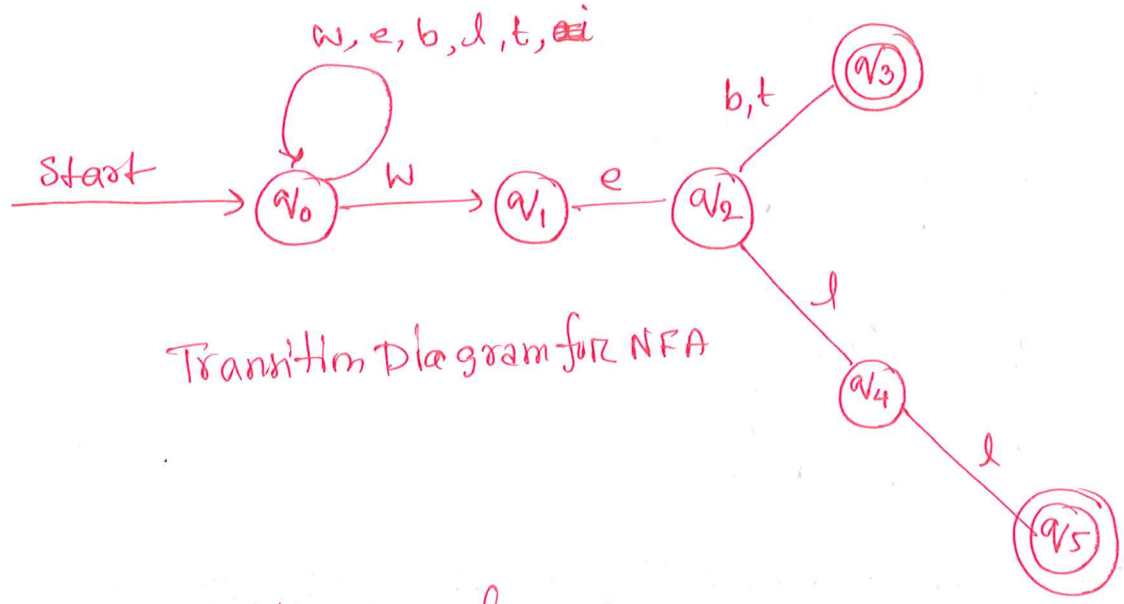
Web, well, wet, wait

b) Convert your NFA to DFA

Solution: a)



So,




Transition Diagram for NFA

Solution

b)

Transition table for DFA

| | w | e | b | d | t |
|---|---|---|---|---|---|
| Start → $\{q_0\}$ | $\{q_0, q_1\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ |
| নতুনটা আসবে $\{q_0, q_1\}$ | $\{q_0 q_1\}$ | $\{q_0, q_2\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ |
| নতুনটা আসবে $\{q_0, q_2\}$ | $\{q_0, q_1\}$ | $\{q_0\}$ | $\{q_0, q_3\}$ | $\{q_0, q_4\}$ | $\{q_3, q_4\}$ |
| নতুনটা আসবে $\{q_0, q_3\}$ * | $\{q_0, q_1\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ |
| নতুনটা আসবে $\{q_0, q_4\}$ | $\{q_0, q_1\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ |
| $\{q_0, q_5\}$ * | $\{q_0 q_1\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_0\}$ |

নিজে করতে হবে।

Remaining the States

$q_0 \xrightarrow{w} q_0 q_1$, $q_1 \xrightarrow{w} \phi$ → $\{q_0, q_1\}$

$q_0 \xrightarrow{e} q_0$, $q_1 \xrightarrow{e} q_2$

$q_0 \xrightarrow{b} q_0$, $q_1 \xrightarrow{b} \phi$ → $q_0$

$q_0 \xrightarrow{d} q_0$, $q_1 \xrightarrow{d} \phi$ → $q_0$

$q_0 \xrightarrow{t} q_0$, $q_1 \xrightarrow{t} \phi$ → $q_0$

$q_0 \xrightarrow{d} q_0$, $q_3 \xrightarrow{d} \phi$ → $q_0$

$q_0 \xrightarrow{w} q_0, q_1$, $q_3 \xrightarrow{w} \phi$ → $q_0, q_1$

$q_0 \xrightarrow{b} q_0$, $q_2 \xrightarrow{b} q_3$ → $q_0, q_3$

$q_0 \xrightarrow{e} q_0$, $q_3 \xrightarrow{e} \phi$ → $q_0$

$q_0 \xrightarrow{d} q_0$, $q_2 \xrightarrow{d} q_4$ → $q_0, q_4$

$q_0 \xrightarrow{t} q_0$, $q_3 \xrightarrow{t} \phi$ → $q_0$

$q_0 \xrightarrow{t} q_0$, $q_2 \xrightarrow{t} q_3$ → $q_0, q_3$

$q_0 \xrightarrow{w} q_0, q_1$, $q_2 \xrightarrow{w} \phi$ → $\{q_0, q_1\}$

$q_0 \xrightarrow{e} q_0$, $q_2 \xrightarrow{e} \phi$ → $q_0$

$q_0 \xrightarrow{w} \{q_0, q_1\}$
$q_4 \xrightarrow{w} \phi$

$q_0, q_1$

$q_0 \xrightarrow{e} q_0$
$q_4 \xrightarrow{e} \phi$

$q_0$

~~ex~~ Final State এলোমেলো * মার্ক দিয়ে চিহ্নিত করা হয়েছে।

Assume:

$\{q_0\} = A$

$\{q_0, q_1\} = B$

$\{q_0, q_2\} = C$

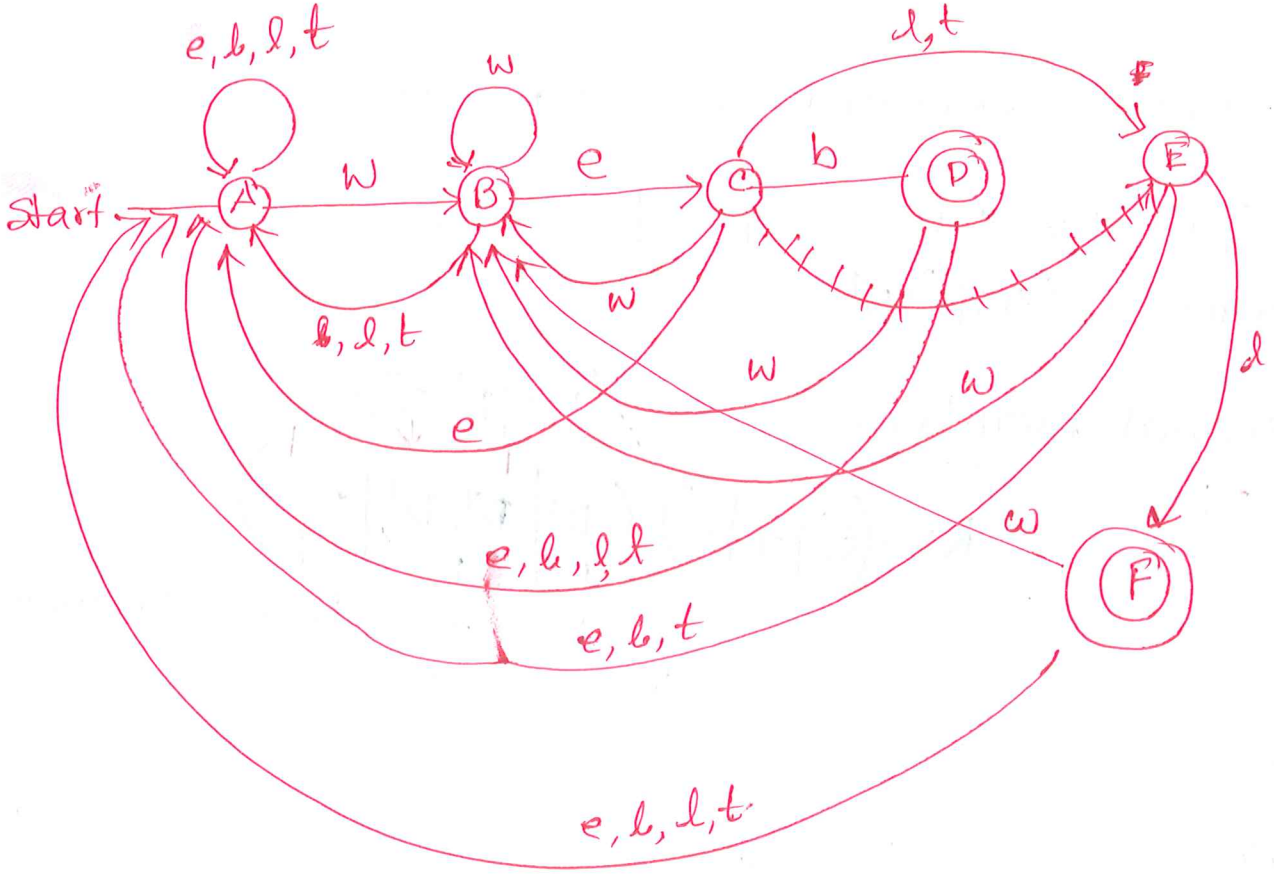
$\{q_0, q_3\} = D$

$\{q_0, q_4\} = E$

$\{q_0, q_5\} = F$

~~DFA~~ Transition table for DFA after renaming

| | w | e | b | d | t |
|---|---|---|---|---|---|
| → A | B | A | A | A | A |
| B | B | C | A | A | A |
| C | B | A | D | E | E |
| * D | B | A | A | A | A |
| E | B | A | A | F | A |
| * F | B | A | A | A | A |




Date: 25/07/2026 (3rd class)

CT: Next class 1/8/2026 starting class to Regular Expression

## Regular Expression

$R^* = \{\epsilon, R, RR, RRR, RRRR \ldots\}$

$R^+ = \{R, RR, RRR, RRRR \ldots\}$

$R_1 | R_2 = R_1 \cup R_2 = R_2 | R_1$

$R_1 . R_2 \neq R_2 . R_1$

$R^* = R^+ \cup \{\epsilon\}$

* ϵ = কিছু না হওয়া সত্ত্বেও

### Integer number

ϵ 2 9 3 — ϵ 1 — ϵ 0

− 9 3 — + 9 2

* ϵ mean কিছুই না

$(\epsilon | + | -) DD^*$

$D = 0|1|2|3|\ldots|9$

$= \{D, DD, DDD, DDDD \ldots\}$

$= \{1, 23, 293, \ldots\}$

$DD^* = D\{\epsilon, D, DD, DDD, DDDD \ldots\}$

$= D\epsilon, DD, DDD, DDDD, DDDDD \ldots$

সাধারণ Integer number এর Regular Expression-

$RE = (\epsilon|+|-)DD^*$ where $D = 0|1|2|3|\ldots|9$

↓

Regular Expression $= (\epsilon|+|-)D^+$

## All decimal numbers

$RE = (\epsilon|+|-)(D^+|D^+.D^+|.D^+)$

$(D^+|D^*.D^+)$

$D^+$ → Integer

$D^+.D^+$ → Fractional

$.D^+$ → fractional numbers < 1

$D^*.D^+$

$= \{\epsilon.D, \epsilon.DD, \ldots\ldots D.DD, DD.DDD\ldots\}$

$= .5, .53 \ldots. 1.23, 24.564$

একটিমাত্র ডিজিট

1

23

293

-93

+92

~~29.~~

একটিমাত্র ডিজিটের পরে ডেসিমাল পয়েন্ট

1.0

29.32

+9.3

-10.4

$D^+.D^+$

একটিমাত্র ডেসিমাল পয়েন্টের পরে

.53

+.53

-.53

$.D^+$

⋮

## GRAMEEN PHONE NUMBER - RE

017 [1 0 2 3 0 0 0 0 0 0]

013 [0 3 4 8 0 0 0 0 0 0]

RE = 01 (3|7) DDDDDDDD, where D = 0|1|2|3|4|5|6|7|8|9

MSC ID - RE — Trimester (Spring-1, Summer-2, Fall-3)

012 232 0023 — Roll no (must not be zero)

MSC Code — Admission year

RE = 012 DD(1|2|3) {(DDD

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RE for (যে সালে ভাবে পড়ে / যে পড়ে) $R_1$

(বহিম পড়ে / তাহার বই পড়ে / বহিম বই পড় / তাহার পড়ে) $R_2$

আমি ভাত খাই
আমরা ভাত খাই
আমি মাছ খাই
আমরা মাছ খাই
আমি ভাত খাইনা
আমি মাছ খাইনা
আমি খাই
আমরা খাই

RE = $R_1 | R_2$

$R_1$ = যে (সালে ভাবে | ∈) পড়ে

$R_2$ = (বহিম | তাহার) (বই | ∈) পড়ে

RE = (আমি | আমরা) (ভাত | মাছ | ∈) খাই (না | ∈)

# GRAMMER

RE = (আমি/আমরা) (ভাত/মাছ/ϵ) খাই (না/ϵ)

V NEG

Subject Object VP

Start → Sub obj Vp

Sub → আমি/আমরা

obj → ভাত/মাছ/ϵ

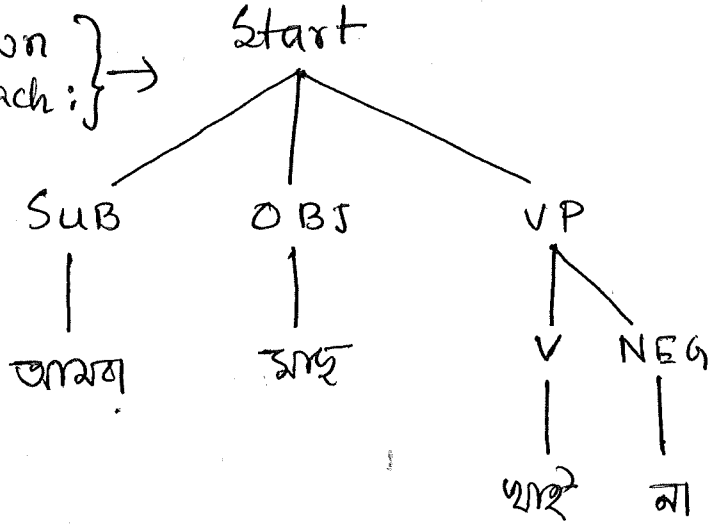
VP → V NEG

V → খাই

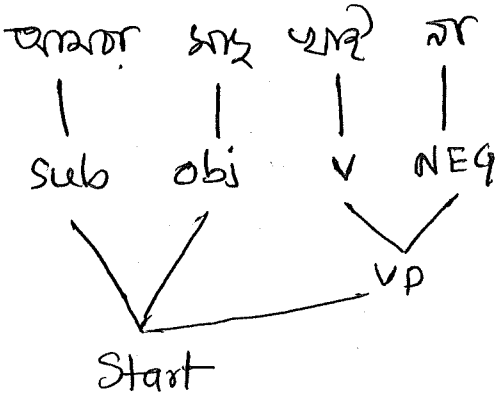
NEG → না/ϵ

In Exam: check whether "আমরা মাছ-খাই না" is a valid String for given grammer

Top down approach: →




Bottom-up approach: →



## Left Most Derivation

Start → Sub Obj VP

→ আমরা Obj VP [Sub → আমরা]

→ আমরা ভাত VP [Obj → ভাত]

→ আমরা ভাত V NEG [VP → V NEG]

→ আমরা ভাত খাই NEG [V → খাই]

→ আমরা ভাত খাই না [NEG → না]

## RIGHT MOST DERIVATION

Start → Sub Obj VP

→ Sub Obj V NEG [VP → V NEG]

⇒ Sub Obj V না [NEG → না]

→ Sub Obj খাই না [V → খাই]

→ Sub ভাত খাই না [Obj → ভাত]

→ আমরা ভাত খাই না [Sub → আমরা]

# SHIFT REDUCE PARSER

Check whether "আমরা ভাত খাই না" is a Valid String for given grammer using shift reduce parser

| Input Buffer | Stack | Parsing Action |
|---|---|---|
| আমরা ভাত খাই না | $ | — |
| ভাত খাই না | $ আমরা | shift |
| খাই না | $ Sub | Reduce by Sub → আমরা |
| না | $ Sub ভাত | shift |
| | $ Sub OBJ | Reduce by OBJ → ভাত |
| | $ Sub OBJ খাই | Shift |
| | $ Sub OBJ V | Reduce by V → খাই |
| | $ Sub OBJ V না | shift |
| | $ Sub OBJ V NEG | Reduce by NEG → না |
| | $ Sub OBJ VP | Reduce by VP → V NEG |
| | $ Start | Reduce by Start → Sub OBJ VP |
| | $ Start ↳ Start | Accept |

Stack is a LIFO Data Structure

$ = Bottom of Stack

Class:-

- Regular Expression from a Language
- Grammer Writing
- String validation by parse tree (Top-down or bottom-up
- Left most derivation / Right most derivation
- String validation by using shift reduce parser

→ ×

Exam-Q: Sum = a+b
Sum = Sum - a
Sum = a + 9
Sum = 9 - b

a) Draw a DFA over alphabet {a, b, ... 0 ... 9}
b) Write regular Expression

letter digit
{ L, D, +, - }
{ (a-z), (0-9) }

